Unified Latent Relation Modeling for Spatio-Temporal Graphs via Language-Model-Driven Global Reasoning

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Abstract

Spatiotemporal graphs are a key abstraction for professional real-world systems characterized by complex dynamics, real-time constraints, and stringent reliability requirements. However, most existing spatiotemporal graph models decompose spatial and temporal dependencies into separate, serial stages. As a result, they often fail to capture long-range cross-node and cross-temporal interactions, limiting global spatiotemporal reasoning. To address this limitation, we propose a unified latent relation framework for spatiotemporal graph learning. It is built on a Latent Relation Adapter (LR-Adapter) that includes a Latent Relation Unifier (LR-Unifier) and a Latent Relation Transformer (LR-Former). LR-Unifier converts temporal displacement into latent relation semantics and constructs a unified latent relation space. LR-Former then performs structured local-to-global aggregation over the resulting latent relation embeddings. We further incorporate these latent relation embeddings into a pretrained LLM via prompt-guided interfaces, leveraging its long-context attention and semantic priors for compositional global reasoning. Our method consistently outperforms strong baselines on six real-world datasets, achieving an average relative gain of 13.69% across all reported metrics and a peak improvement of 33.85%, which demonstrates the effectiveness of unified latent relation modeling with language-model-driven global reasoning.

CCS Concepts

• **Computing methodologies → Neural networks**; *Natural language processing*; Knowledge representation and reasoning.

Keywords

Latent Relation Modeling, Temporal Graph, Spatio-Temporal Prediction, Large Language Model

1 Introduction

Spatiotemporal graphs are pervasive in real-world systems where entities evolve and interact over both space and time, such as social interaction analysis [6], recommendation systems [35], traffic networks [23, 28], epidemic spreading [43], and infrastructure monitoring [10, 51]. A core challenge in spatiotemporal graph learning is to capture dependencies spanning multiple nodes, time steps, and relation types [14, 15], as these dependencies are essential for prediction, reasoning, and decision-making in complex dynamic environments yet remain difficult to model effectively [16, 18, 38].

A fundamental limitation of existing spatiotemporal graph models is their stage-wise factorization of joint dependencies. By modeling spatial and temporal interactions in separate, serial stages, most methods impose a separability bias on intrinsically coupled spatiotemporal processes [7, 12, 15, 21, 24, 53]. As a result, explicit priors for long-range cross-temporal and cross-node interactions are obscured, and such dependencies cannot be directly encoded as unified relations, but must instead be approximated through sequential multi-step propagation [27]. Although effective for local information aggregation, this indirect approximation restricts long-range dependency modeling and suppresses the potential for global spatiotemporal reasoning.

Recent studies have attempted to mitigate this limitation from three directions. First, some methods replace stage-wise spatial-temporal pipelines with more direct interaction modeling, e.g., transformer-style attention over graph sequences or learnable long-range spatiotemporal graphs, to allow cross-time and cross-location information exchange [21, 45]. Second, other works attempt to improve long-range dependency modeling by adopting alternative temporal backbones, such as state space models and continuous-time propagation [25], motivated by the limitations of recurrent, message-passing, and self-attention architectures in capturing long-range temporal dependencies in graph tasks [11, 27]. Third, LLM-enhanced methods explore whether pretrained language models can provide stronger semantic priors for spatiotemporal graph learning [50]. However, these methods still fall short of explicitly representing temporal displacement as relational semantics, and they do not fully address global reasoning over unified, densely entangled spatiotemporal relations. These limitations highlight two fundamental challenges. First, spatiotemporal dependencies are not explicitly modeled in a relation-centric manner, but are instead decomposed hierarchically into spatial and temporal components. Second, even with unified representations, existing models remain limited in performing effective global reasoning over densely entangled relations across nodes, time steps, and contexts.

In this work, we propose a unified latent relation framework for spatiotemporal graph learning. Specifically, we introduce a *Latent Relation Adapter (LR-Adapter)* that consists of *Latent Relation Unification (LR-Unifier)* and a *Latent Relation Transformer (LR-Former)*. LR-Unifier maps temporal displacement into latent relations and reformulates spatiotemporal interactions in a shared latent relation space, while LR-Former performs structured local-to-global aggregation over this space. Built on this representation, we further couple the LR-Adapter with a pretrained large language model, where the adapter captures structured dependencies and the language model performs compositional global reasoning. In this way, our framework addresses both the structural limitation of hierarchical decomposition and the reasoning limitation of conventional graph encoders, achieving state-of-the-art performance across multiple spatiotemporal graph learning tasks on six real-world datasets. Our contributions are summarized as follows:

• We present a unified latent relation framework for spatiotemporal graph learning. To the best of our knowledge, this is the first framework that enables global reasoning over unified latent relations, addressing the limitation of hierarchical factorization that reduces explicit joint interactions to indirect sequential approximations and weakens long-range reasoning.

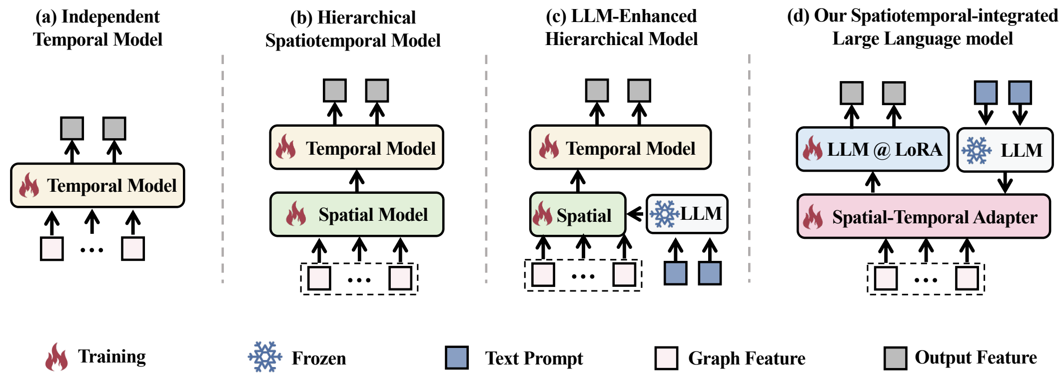

Figure 1: Four types of spatiotemporal models: (a) independent temporal models; (b) hierarchical spatiotemporal models; (c) LLM-enhanced hierarchical spatiotemporal models; and (d) our spatiotemporal-integrated large language model.

• We propose a *Latent Relation Adapter (LR-Adapter)* composed of *Latent Relation Unification (LR-Unifier)* and a *Latent Relation Transformer (LR-Former)*. LR-Unifier maps temporal displacement into latent relations and enables explicit modeling of cross-time and cross-node dependencies in a unified latent relation space, while LR-Former performs structured local-to-global aggregation over this space. Coupled with a pretrained LLM, the framework further enables compositional global reasoning over unified latent relations.

• Extensive experiments on six real-world datasets demonstrate the effectiveness and robustness of our method across multiple tasks, particularly under few-shot and zero-shot settings. We further release a large-scale mine pressure dataset as a new benchmark for spatiotemporal forecasting and risk analysis.

2 Related Work

Independent Temporal Models. As illustrated in Figure 1(a), many forecasting methods focus on temporal dynamics without explicitly modeling inter-node relations, including ARIMA [2], multilayer perceptrons [48], CNN-based models [4], recurrent networks such as LSTM [17], and Transformer-based forecasting models with independently encoded temporal features [31]. In contrast, our work targets spatiotemporal graphs and jointly models dependencies across nodes, time steps, and relation types.

Spatiotemporal Hierarchical Models. A mainstream paradigm in spatiotemporal graph learning is to model spatial and temporal dependencies in a hierarchical manner, as shown in Figure 1(b). Representative graph encoders include GCN [22], GAT [40], R-GCN [37], and HGT [19]. Based on them, many forecasting models have been proposed, such as STGCN [49], AGCRN [1], MTGNN [46], DySAT [36], and HTGNN [8, 9]. Different from these hierarchical designs, our method reformulates spatiotemporal interactions in a unified relation-centric space.

LLM-Enhanced Spatiotemporal Hierarchical Models. Recent studies incorporate pretrained language models into spatiotemporal learning as semantic priors, as shown in Figure 1(c). In these methods, LLMs or PLMs mainly serve as external semantic enhancers for downstream spatiotemporal models [33, 41], while the core modeling process remains hierarchical, with spatial and temporal dependencies handled by conventional graph and temporal modules [44, 52]. In contrast, our framework integrates the language model into reasoning over unified spatiotemporal relations, rather than using it only as a prior injector.

Spatiotemporal-Integrated Large Language Models. More recent works move beyond prior injection and directly develop LLM-based frameworks for spatiotemporal forecasting, as shown in Figure 1(d). Existing methods mainly adopt two strategies: hierarchical spatiotemporal adapters [32] or token-level spatiotemporal fusion before LLM reasoning [30]. In contrast, our method introduces a relation-centric Latent Relation Adapter that first unifies spatial and temporal dependencies into a shared relation space, allowing the LLM to reason globally over the unified representation. This makes our framework different from both hierarchical-adapter and token-fusion designs.

3 Preliminaries

This section presents preliminaries of spatiotemporal graphs.

Definition 1 (Graph). A graph is defined as $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ with node-type mapping $\phi_n : \mathcal{V} \to C_n$ and edge-type mapping $\phi_e : \mathcal{E} \to C_e$, where C_n and C_e denote the sets of node types and edge types, respectively. If $|C_n| = 1$ and $|C_e| = 1$, the graph is homogeneous; otherwise, it is heterogeneous.

Definition 2 (Temporal Graph). A temporal graph is a sequence of evolving graph snapshots, denoted by $\mathcal{G} = \{\mathcal{G}_t\}_{t=1}^{T}$, where $\mathcal{G}_t = (\mathcal{V}_t, \mathcal{E}_t)$ is the graph at time step t. The mappings ϕ_n and ϕ_e are shared across timestamps, and the overall node and edge sets are $\mathcal{V} = \bigcup_{t=1}^{T} \mathcal{V}_t$ and $\mathcal{E} = \bigcup_{t=1}^{T} \mathcal{E}_t$.

Problem Statement. Given a temporal graph $\mathcal{G} = \{\mathcal{G}_t\}_{t=1}^{T}$, the goal is to learn node representations $\mathbf{z}_v^{(T)} \in \mathbb{R}^d$ for each $v \in \mathcal{V}$ from historical snapshots up to time T, such that they capture structural, temporal, and semantic information in the evolving graph. The learned representations are then used for downstream tasks at future time steps, including node regression, link prediction, and node classification.

4 Methodology

Figure 2 illustrates the overall framework of the proposed method, including prompt embeddings, a Latent Relation Adapter (detailed in Figure 3), a generative pre-trained large language model, and an

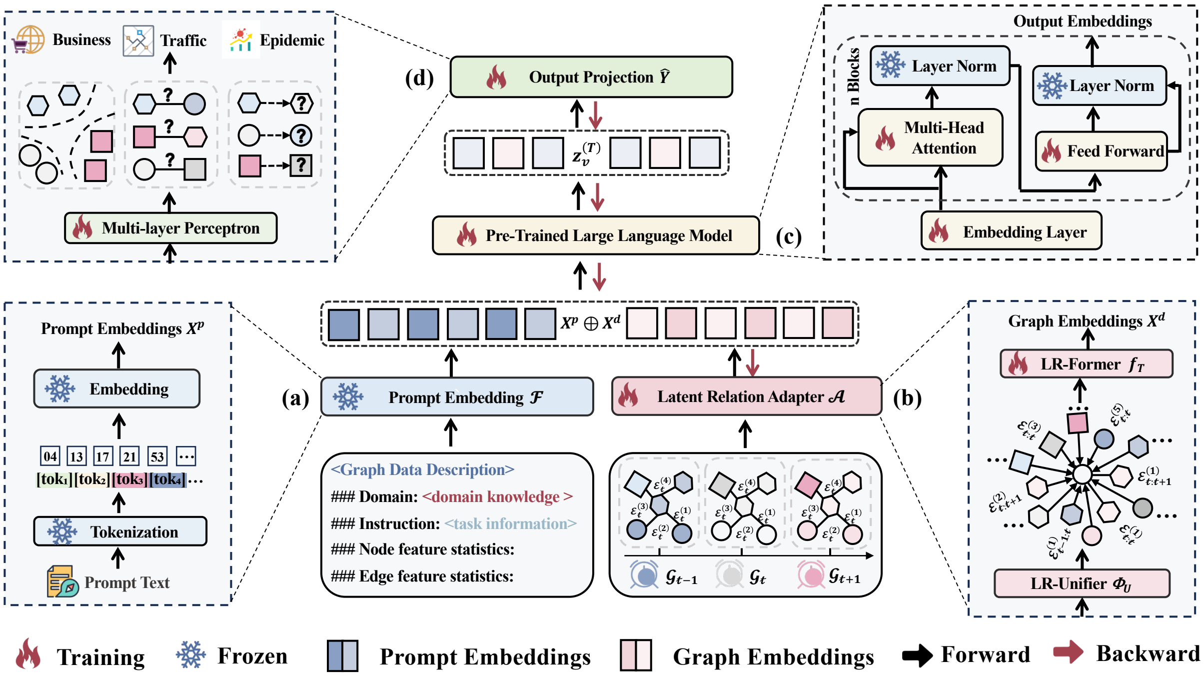

Figure 2: The overall architecture of our method. (a) Prompt embedding of the temporal graph. (b) Latent Relation Adapter. (c): Concatenation of prompt embeddings and temporal graph embeddings, followed by parameter-efficient fine-tuning of the pre-trained large language model. (d) Output projection head for downstream node regression and link prediction tasks.

output projection module. The following subsections describe each component in detail.

4.1 Prompt Embedding

We adopt a prompt strategy to guide the LLM in modeling temporal graph data, consisting of: (1) dataset context, (2) task instruction, and (3) spatio-temporal statistical summaries (Figure 4). The dataset context provides background and natural language descriptions of graph topology, including node/edge types, temporal dynamics, and key properties (e.g., directedness, evolution). The task instruction defines the prediction goal, such as node forecasting or temporal link prediction. Statistical summaries offer key structural and feature-level metrics (e.g., max/min/avg degrees and feature values) to help the model capture essential spatiotemporal patterns.

The spatiotemporal textual embedding is defined as follows:

$$X^p = \mathcal{F}(X_{t-s:t}, D_{t-s:t}), \tag{1}$$

where $X^p \in \mathbb{R}^{P \times H}$, P is the embedding length, and H denotes the hidden size of the LLM. $X_{t-s:t} \in \mathbb{R}^{(s+1) \times N \times C}$ denotes the node features from time step $t-s$ to t, while $D_{t-s:t} \in \mathbb{R}^{(s+1) \times N}$ represents the corresponding node degrees. The mapping $\mathcal{F}(\cdot)$ is not a simple concatenation, but a template-based slot-filling procedure that converts structured spatiotemporal graph statistics into natural-language descriptions.

4.2 Latent Relation Adapter

To overcome the structural limitation of hierarchical spatiotemporal modeling, we introduce a *Latent Relation Adapter* that transforms a temporal graph sequence into unified spatiotemporal relational embeddings for downstream reasoning. Given a temporal graph sequence $\{\mathcal{G}_t\}_{t=1}^{T_w}$, the adapter is defined as

$$\mathbf{X}^d = \mathcal{A}\Big(\{\mathcal{G}_t\}_{t=1}^{T_w}\Big) = f_{\mathrm{T}}\Big(\Phi_{\mathrm{U}}\Big(\{\mathcal{G}_t\}_{t=1}^{T_w}\Big)\Big), \tag{2}$$

where $\Phi_{\mathrm{U}}(\cdot)$ denotes *Latent Relation Unifier (LR-Unifier)* and $f_{\mathrm{T}}(\cdot)$ denotes the *Latent Relation Transformer (LR-Former)*. LR-Unifier lifts temporal displacement into relational semantics to construct a unified spatiotemporal relational graph, while LR-Former performs structured local-to-global reasoning over the unified graph and produces LLM-compatible embeddings $\mathbf{X}^d$.

Latent Relation Unifier. Most existing spatiotemporal graph learning methods model temporal dynamics by attaching time encodings to nodes or edges, or by stacking temporal modules on top of spatial graph operators (Figure 3(a)). Such designs treat time as an auxiliary signal rather than an intrinsic component of relational structure, making it difficult to explicitly model delayed or cross-temporal interactions. To address this issue, LR-Unifier defines an atomic spatiotemporal relation as

$$(r, \tau) \in \mathcal{R} \times \mathcal{T}, \tag{3}$$

where r denotes a semantic relation type and τ denotes the temporal offset between two graph snapshots. This formulation incorporates temporal displacement into relation types, thereby converting the original temporal graph sequence into the following unified spatiotemporal relational graph:

$$\tilde{\mathcal{G}} = \Phi_{\mathrm{U}}\Big(\{\mathcal{G}_t\}_{t=1}^{T_w}\Big) = (\tilde{\mathcal{V}}, \tilde{\mathcal{E}}, \tilde{\mathcal{R}}), \tag{4}$$

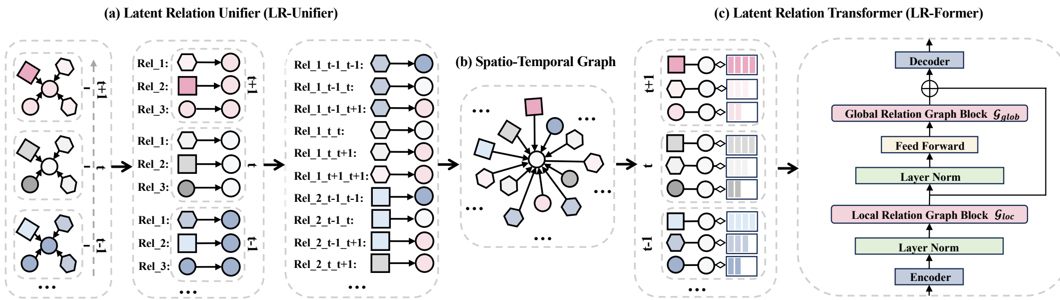

Figure 3: Latent Relation Adapter. (a) Latent Relation Unifier (LR-Unifier). (b) Spatio-Temporal Graph. (c) Latent Relation Transformer (LR-Former).

where each edge type in $\tilde{\mathcal{R}}$ corresponds to a spatiotemporal relation (r, τ). Formally, for nodes i and j connected by relation r at timestamps t and t', LR-Unifier introduces a directed edge

$$(i^{(t)}, j^{(t')}) \xrightarrow{(r,\tau)}, \qquad \tau = t' - t. \tag{5}$$

By enumerating relation types and temporal offsets within the temporal window T_w, LR-Unifier reformulates temporal modeling as relational reasoning on a higher-order graph, thereby enabling direct cross-temporal interactions (Figure 3(b)). From a propagation perspective, hierarchical temporal models require $O(\tau)$ sequential message-passing steps to transmit information across τ timestamps, whereas LR-Unifier reduces the effective propagation path length to $O(1)$ in a single relational step.

Latent Relation Space Transformer. Based on the unified spatiotemporal relational graph $\tilde{\mathcal{G}}$, we design a *Latent Relation Transformer (LR-Former)* to extract compact relational embeddings while maintaining compatibility with large language models. As illustrated in Figure 3(c), LR-Former follows a structured local-to-global pipeline consisting of a lightweight encoder, a local relational module, a global relational module, and a decoder. The encoder and decoder are implemented as feed-forward projections, while the local and global modules are instantiated by abstract relation-aware graph operators.

Let $\mathbf{X}$ denote the input feature matrix to LR-Former on the unified spatiotemporal relational graph. The forward process is defined as

$$\mathbf{X}^{e} = \mathrm{Enc}(\mathbf{X}), \tag{6}$$

$$\mathbf{X}_{\mathrm{loc}} = \mathcal{G}_{\mathrm{loc}}(\mathrm{LN}(\mathbf{X}^{e}))\,, \tag{7}$$

$$\mathbf{X}_{\mathrm{glob}} = \mathcal{G}_{\mathrm{glob}}(\mathrm{FFN}(\mathrm{LN}(\mathbf{X}_{\mathrm{loc}})))\,, \tag{8}$$

$$\mathbf{X}^{d} = \mathrm{Dec}\big(\mathbf{X}_{\mathrm{loc}} + \mathbf{X}_{\mathrm{glob}}\big)\,, \tag{9}$$

where $\mathbf{X}^{e}$ denotes the encoded hidden representation, $\mathbf{X}_{\mathrm{loc}}$ and $\mathbf{X}_{\mathrm{glob}}$ denote the local and global relational features, respectively, and $\mathbf{X}^{d}$ is the final adapter output projected into the LLM language embedding space.

The encoder first maps unified spatiotemporal node features into a shared latent space, making representations from different timestamps and relation contexts comparable. The local operator $\mathcal{G}_{\mathrm{loc}}(\cdot)$ then captures fine-grained spatiotemporal interactions within local relational neighborhoods, emphasizing short-range dependencies and relation-aware structural patterns. On top of the local relational features, the global operator $\mathcal{G}_{\mathrm{glob}}(\cdot)$ further models long-range dependencies over the transformed representation, enabling information integration across distant nodes, relation types, and temporal offsets. Finally, the decoder projects the fused local-global representation into the hidden space of the large language model, so that the resulting graph embeddings can be jointly processed with prompt embeddings in subsequent reasoning.

Relational Graph Block. Both the local and global modules adopt the same *Relational Graph Block*, as illustrated in Figure 5. The block follows a Transformer-style architecture consisting of layer normalization, a relation-aware graph operator, and a feed-forward network with residual connections. Let $\mathbf{H}^{(l)}$ denote the input node representations at layer l. The block first applies pre-normalization and a graph operator, followed by a residual connection:

$$\tilde{\mathbf{H}}^{(l)} = \mathbf{H}^{(l)} + \mathcal{G}\Big(\mathrm{LN}(\mathbf{H}^{(l)})\Big)\,, \tag{10}$$

where $\mathcal{G}(\cdot)$ denotes a *relation-aware graph operator*. A second normalization and feed-forward transformation are then applied:

$$\mathbf{H}^{(l+1)} = \tilde{\mathbf{H}}^{(l)} + \mathrm{FFN}\Big(\mathrm{LN}(\tilde{\mathbf{H}}^{(l)})\Big)\,. \tag{11}$$

This design stabilizes deep relational propagation while preserving the expressiveness of graph-based message passing.

The operator $\mathcal{G}(\cdot)$ is designed to be *plug-and-play* and can be instantiated with different relational graph operators, such as relation-aware graph convolution or relation-aware graph attention. Here, we use *relation-aware graph attention* as an example. Following the standard *score–normalize–aggregate* paradigm of graph attention [3], but replacing conventional neighborhood propagation with our unified spatiotemporal relational aggregation, we define the attention score for edge (j, i) under semantic relation r and temporal offset τ as

$$e_{ij}^{(r,\tau,l)} = \mathrm{LeakyReLU}\Big(\mathbf{a}^{\top}\big[\mathbf{W}_r^{(l)}\mathbf{h}_i^{(t,l)} \,\|\, \mathbf{W}_r^{(l)}\mathbf{h}_j^{(t-\tau,l)}\big]\Big)\,, \tag{12}$$

where $\mathbf{a}$ is a learnable attention vector and $\|$ denotes vector concatenation.

The corresponding attention coefficients are normalized within the spatiotemporal neighborhood, ensuring that the contributions of neighboring nodes are properly scaled across different semantic relations and temporal offsets. This normalization allows the model to selectively emphasize the most informative dependencies while

[START PROMPT]

[Dataset Description]:
The COVID-19 dataset captures the daily progression of the epidemic in the United States, including both state-level and county-level case reports.

[Graph Task]:
Forecast the values of <node_num> <node_type> in the next time step, based on observations from the previous <history_time> steps.

[Graph Topology Statistics]:
The temporal graph exhibits a minimum node degree of <*min_degree*>, a maximum degree of <*max_degree*>, and a median degree of <*avg_degree*>.

[Node Feature Statistics]:
Minimum feature value: <*min_feature*>; Maximum feature value: <*max_feature*>; Median feature value: <*median_feature*>.

[END PROMPT]

Figure 4: Prompt template.

suppressing less relevant cross-time interactions during message aggregation.

$$\alpha_{ij}^{(r,\tau,l)} = \mathrm{softmax}_j\left(e_{ij}^{(r,\tau,l)}\right) = \frac{\exp\left(e_{ij}^{(r,\tau,l)}\right)}{\sum_{j'\in\mathcal{N}_i^{(r,\tau)}} \exp\left(e_{ij'}^{(r,\tau,l)}\right)}. \tag{13}$$

The updated representation of node i is then computed by unified spatiotemporal relational propagation:

$$\mathbf{h}_i^{(t,l+1)} = \sigma\Bigg(\sum_{r\in\mathcal{R}}\sum_{\tau\in\mathcal{T}}\sum_{j\in\mathcal{N}_i^{(r,\tau)}} \alpha_{ij}^{(r,\tau,l)}\,\mathbf{W}_r^{(l)}\mathbf{T}_\tau^{(l)}\mathbf{h}_j^{(t-\tau,l)} + \mathbf{W}_0^{(l)}\mathbf{h}_i^{(t,l)}\Bigg), \tag{14}$$

where $\mathbf{h}_i^{(t,l)} \in \mathbb{R}^d$ denotes the representation of node i at spatiotemporal step t in layer l, $\mathcal{N}_i^{(r,\tau)}$ is the neighbor set associated with relation r and offset τ, $\mathbf{W}_r^{(l)}$ and $\mathbf{T}_\tau^{(l)}$ are relation-specific and offset-specific transformations, $\mathbf{W}_0^{(l)}$ models self-relational preservation, and $\sigma(\cdot)$ is a non-linear activation. The coefficient $\alpha_{ij}^{(r,\tau,l)}$ determines how messages are weighted under relation type r and temporal offset τ.

Compared with standard graph attention, the key difference is that the neighborhood aggregation is no longer defined over ordinary graph edges, but over the unified spatiotemporal relation space induced by LR-Unifier [3]. As a result, the block can adaptively weight neighbors across both semantic relations and temporal displacements, enabling explicit modeling of heterogeneous cross-temporal dependencies within a single relational operator.

4.3 Pretrained Large Language Model

We employ a decoder-only generative pre-trained large language model as the backbone for spatiotemporal representation learning (Figure 2(c)) and experiment with multiple LLM backbones. The input sequence is constructed by concatenating the prompt embeddings and the spatiotemporally fused relational embeddings, denoted as $\mathbf{X}^p \oplus \mathbf{X}^d$. The resulting embeddings are projected to Q, K, and V and processed by self-attention:

$$Z = Attention(Q,K,V) = softmax\left(\frac{QK^\top}{\sqrt{d_k}}\right)V. \tag{15}$$

We use LoRA for parameter-efficient adaptation. For ultra-large spatiotemporal graphs, the compression mechanism of our Latent Relation Adapter, together with recent long-context techniques, enables more efficient modeling of large serialized spatiotemporal contexts [34].

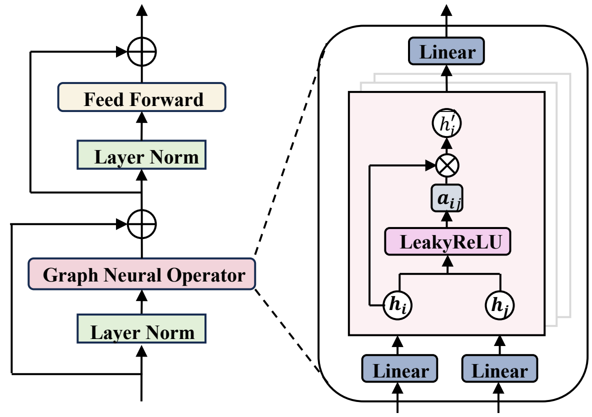

Figure 5: The Architecture of Relational Graph Block.

Table 1: Summary Statistics of the Datasets. NR: Node Regression; NC: Node Classification; LP: Link Prediction.

Dataset	Task	Nodes	Edges	Snapshot	Win.	Hor.
CHBike	NR	250	106,996	30 min	12	12
Pressure	NR	360	4,186	1 hour	12	12
SZFire	NR	1200	10,000	1 sec	12	12
Enron	LP	184	1,690	1 day	12	1
COVID-19	NR	3,277	22,176	1 day	7	1
Yelp	NC	68,226	125,235	1 month	12	1

4.4 Output Projection

Let $\mathbf{H}$ denote the hidden states produced by the fine-tuned large language model for the input sequence $\mathbf{X}^p \oplus \mathbf{X}^d$. The final predictions are obtained by applying a multi-layer perceptron to the hidden states corresponding to the spatiotemporal tokens after removing the prefix part:

$$\widehat{\mathrm{Y}} = \mathrm{MLP}\left(\mathbf{H}_{L_p+1:L_p+L_d}\right), \tag{16}$$

where L_p and L_d denote the lengths of the prompt tokens and spatiotemporal tokens, respectively, as illustrated in Figure 2(d).

5 Experiments

In this section, we evaluate the performance of our method on a series of experiments over several real-world datasets.

5.1 Datasets and Task Setup

We evaluate our method on six real-world datasets: CHBike [33], Mine pressure, SZFire, COVID-19 [41], Enron [36], and Yelp [20],

Table 2: Performance comparison on Mine Pressure, CHBike, and SZFire datasets. U-Rel.: explicit unified relational modeling of temporal displacement; Joint: single-space spatiotemporal modeling rather than stage-wise decomposition; Global: explicit non-local/global reasoning beyond local message passing. Avg. Rank is averaged over the six reported metrics.

Method	U-Rel.	Joint	Global	Mine Pressure		CHBike		SZFire		Avg. Rank (↓)
				MAE (↓)	RMSE (↓)	MAE (↓)	RMSE (↓)	MAE (↓)	RMSE (↓)	
DCRNN	×	×	×	3.32±0.06	5.80±0.11	1.95±0.12	2.97±0.15	3.75±0.14	16.85±0.25	9.75
STGCN	×	×	×	3.23±0.09	5.76±0.21	2.12±0.23	3.11±0.24	3.72±0.10	17.73±0.28	10.50
ASTGCN	×	×	×	3.15±0.11	5.45±0.09	2.83±0.15	4.29±0.19	3.68±0.12	21.65±0.27	11.50
GWN	×	×	×	3.29±0.13	5.78±0.17	1.94±0.12	3.04±0.06	3.66±0.19	18.60±0.26	9.33
AGCRN	×	×	×	3.18±0.05	5.75±0.12	2.06±0.23	3.21±0.12	3.63±0.09	17.55±0.25	8.75
GMAN	×	×	✓	3.94±0.19	5.32±0.08	2.11±0.16	3.09±0.24	3.77±0.12	19.95±0.30	11.67
STSGCN	×	✓	×	3.18±0.08	5.25±0.05	2.43±0.02	4.50±0.26	3.62±0.18	22.52±0.24	10.75
ASTGNN	×	×	✓	3.20±0.14	5.27±0.11	2.23±0.04	3.48±0.13	3.70±0.11	18.70±0.28	10.83
DGCRN	×	×	×	3.46±0.05	5.20±0.13	1.95±0.15	2.97±0.19	3.73±0.10	17.80±0.27	8.92
OFA	×	×	✓	2.64±0.12	4.57±0.16	1.95±0.12	2.95±0.21	3.97±0.11	18.30±0.25	8.00
GATGPT	×	×	✓	2.62±0.08	4.55±0.13	1.96±0.11	2.94±0.20	3.86±0.13	15.28±0.24	6.83
GCNGPT	×	×	✓	2.60±0.07	4.52±0.20	2.25±0.14	3.42±0.39	3.65±0.15	14.26±0.23	7.17
ST-LLM	×	×	✓	2.57±0.09	4.48±0.11	1.91±0.02	2.90±0.07	3.35±0.07	13.25±0.18	2.67
ST-LLM+	×	×	✓	2.59±0.14	4.49±0.16	1.89±0.12	2.79±0.18	3.24±0.14	12.24±0.16	2.33
Ours	✓	✓	✓	**1.70±0.13**	**3.08±0.21**	**1.75±0.12**	**2.68±0.14**	**2.50±0.16**	**9.60±0.23**	**1.00**
Improv.				+33.85%	+31.25%	+7.41%	+3.94%	+22.84%	+21.57%	+57.08%

Table 3: Performance comparison on COVID-19, Enron, and Yelp datasets. U-Rel.: explicit unified relational modeling of temporal displacement; Joint: single-space spatiotemporal modeling rather than stage-wise decomposition; Global: explicit non-local/global reasoning beyond local message passing. Avg. Rank is averaged over the six reported metrics.

Method	U-Rel.	Joint	Global	COVID-19		Enron		Yelp		Avg. Rank (↓)
				MAE (↓)	RMSE (↓)	AUC (%) ↑	AP (%) ↑	F1 (%) ↑	Recall (%) ↑	
GCN	×	×	×	834±64	1476±59	74.02±1.23	75.83±0.94	37.18±0.55	38.12±0.76	10.75
GAT	×	×	×	728±24	1565±79	71.65±0.34	76.33±0.91	35.42±1.17	35.58±1.41	12.00
RGCN	×	×	×	823±94	1553±97	71.57±1.19	73.02±1.21	37.83±0.92	40.18±0.51	10.67
HAN	×	×	×	692±24	1491±85	71.48±0.61	72.23±1.12	36.90±1.18	39.13±1.70	11.17
HGT	×	×	×	643±19	1382±47	75.80±0.41	75.41±0.53	40.32±1.26	40.69±0.54	7.83
DySAT	×	×	✓	565±09	1219±17	82.48±0.98	81.31±0.91	37.83±0.64	38.17±0.76	7.17
DyHATR	×	×	×	760±12	1489±39	80.51±0.23	81.98±0.24	36.34±0.51	40.56±1.45	9.33
HGT+	×	×	×	-	-	86.51±0.72	87.98±0.95	37.83±0.64	38.12±0.76	$5.62^{\dagger}$
HTGNN	×	×	×	550±29	1137±32	83.67±0.51	84.34±1.04	40.45±1.45	40.28±0.59	4.67
RGCN-LLM	×	×	✓	576±43	1239±64	84.31±0.45	83.95±0.66	37.45±0.70	36.72±0.85	7.33
RGAT-LLM	×	×	✓	598±56	1349±89	83.03±0.61	82.38±0.92	36.53±0.64	34.67±0.26	9.00
CasMLN	×	×	×	543±10	1115±56	85.24±0.53	84.38±0.42	42.86±0.64	42.67±0.76	3.17
HTGformer	×	×	✓	533±23	1215±44	87.37±0.35	86.59±0.50	43.34±0.90	43.87±0.54	2.50
Ours	✓	✓	✓	**494±21**	**1018±13**	**93.71±0.57**	**94.30±0.78**	**45.68±0.21**	**46.61±0.95**	**1.00**
Improv.				+9.02%	+8.70%	+6.34%	+7.71%	+5.40%	+6.26%	+60.00%

covering node regression, link prediction, and node classification. Their statistics are summarized in Table 1. Specifically, CHBike, Mine pressure, SZFire, and COVID-19 are used for node regression, Enron for link prediction, and Yelp for node classification. These datasets span diverse application domains, including urban mobility, industrial monitoring, public health, email communication, and online review networks.

5.2 Baselines

We compare our method with over 20 representative baselines, including DCRNN [29], STGCN [49], AGCRN [1], STSGCN [39], ASTGNN [13], DGCRN [26], OFA [54], GATGPT [5], GCNGPT [33], ST-LLM [33], ST-LLM+ [32], RGCN [37], HAN [42], HGT [19], DySAT [36], DyHATR [47], HGT+ [19], HTGNN [8], HTGformer [43], RGCN-LLM, RGAT-LLM, and CasMLN [41].

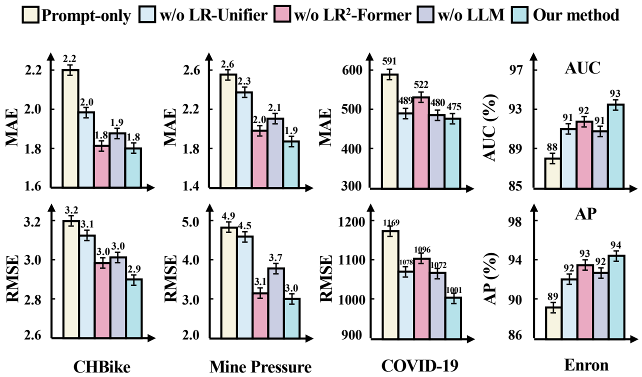

Figure 6: Ablation study of our method.

5.3 Implementation Details

Experiments are conducted on an NVIDIA H800 GPU with 80GB memory. We use Adam with a learning rate of 0.001 and a weight decay of 0.0001. The pretrained large language models include LLaMA-3.2 (1B), GPT-2 (1.5B), LLaMA-2 (7B), and LLaMA-3.1 (8B). Gradient accumulation is adopted to reduce computational overhead, and all models are trained for 500 epochs.

5.4 Overall Performance

Tables 2 and 3 compare our method with state-of-the-art baselines on six real-world datasets. All deep learning models are evaluated with five random seeds, and the mean ± standard deviation is reported. Overall, LLM-enhanced baselines generally outperform earlier spatiotemporal models. For example, on Mine Pressure, ST-LLM improves MAE/RMSE from OFA's 2.64/4.57 to 2.57/4.48, while on SZFire, ST-LLM+ achieves 3.24/12.24 versus GWN's 3.66/18.60. On Table 3, HTGformer is the strongest baseline, obtaining 533/1215 on COVID-19, 87.37/86.59 on Enron, and 43.34/43.87 on Yelp, but it still ranks behind our method overall, with Avg. Rank 2.33 and 2.50 on the two tables.

Our method achieves the best results on all six datasets and all reported metrics. On Table 2, it improves Mine Pressure from 2.57/4.48 to 1.70/3.08, yielding 33.85%/31.25% gains, and reduces SZFire from 3.24/12.24 to 2.50/9.60, corresponding to 22.84%/21.57% gains. On the more competitive CHBike dataset, it still improves from 1.89/2.79 to 1.75/2.68 (7.41%/3.94%). On Table 3, our method reduces COVID-19 from 533/1115 to 494/1018 (9.02%/8.70%), improves Enron from 87.37/86.59 to 93.71/94.30 (6.34%/7.71%), and raises Yelp from 43.34/43.87 to 45.68/46.61 (5.40%/6.26%). Overall, the gains range from 3.94% to 33.85%, with the largest improvements on Mine Pressure, SZFire, and Enron, indicating that unified latent relation modeling with global reasoning is particularly effective for datasets with stronger long-range cross-time and cross-node dependencies.

5.5 Ablation Study

We study four variants: **Prompt-only**, using only textual prompts; **w/o LR-Unifier**, removing Latent Relation Unifier; **w/o LR-Former**, replacing the structured adapter with feed-forward layers; and **w/o LLM**, replacing the pretrained large language model with a single-layer Transformer.

Figure 6 shows that **Prompt-only** performs worst across all datasets, indicating that text alone is insufficient for fine-grained spatiotemporal modeling. Removing **LR-Unifier** or **LR-Former** consistently hurts performance, verifying the importance of unified relation modeling and structured aggregation. Replacing the pretrained **LLM** also leads to clear degradation, confirming the benefit of large-scale pretraining. These results show that LR-Unifier, LR-Former, and the LLM contribute complementary gains to the full model.

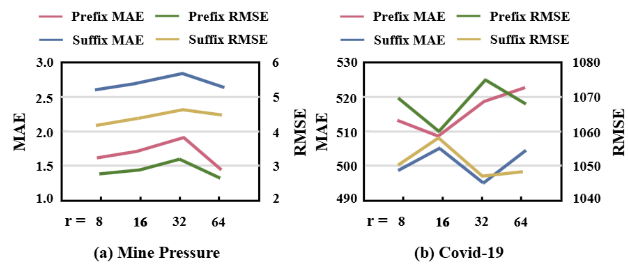

Figure 7: Performance of different LoRA ranks and prompt placements (prefix vs. suffix) for the Mine Pressure (a) and COVID-19 (b) datasets.

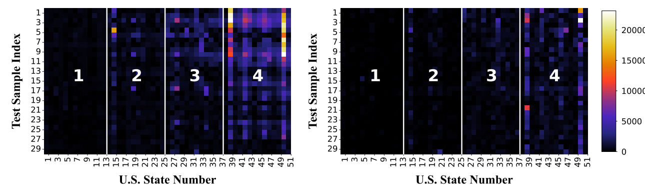

Figure 8: Heatmap visualization of prediction errors. Left: CasMLN. Right: our method. Higher color intensity indicates larger error.

5.6 Effect of Different Parameters

We study two prompt placement strategies, i.e., prefix and suffix (Figure 4). For Mine Pressure (Figure 7(a)), prefix prompts consistently outperform suffix prompts across all LoRA ranks. The best results are obtained at rank 8 (MAE = 1.53, RMSE = 2.80), with a slight further gain at rank 64 (MAE = 1.48, RMSE = 2.73). By contrast, suffix prompts yield consistently larger errors. For COVID-19 (Figure 7(b)), suffix prompts perform best at rank 8 (MAE = 498, RMSE = 1051), while performance slightly degrades at higher ranks. Prefix prompts are less stable, with noticeably increased MAE and RMSE at ranks 32 and 64. Overall, prefix prompts are more suitable for Mine Pressure, whereas suffix prompts are more effective for COVID-19, especially at lower LoRA ranks.

5.7 Case Study

We evaluate robustness on COVID-19 by clustering the 51 U.S. states into four groups based on the median, interquartile range,

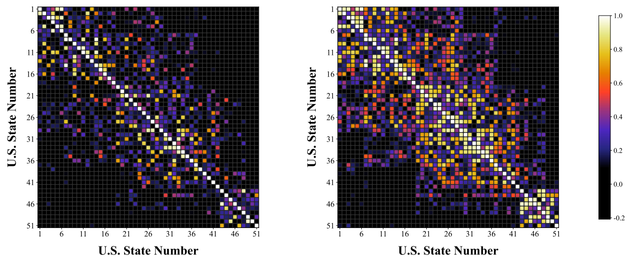

Figure 9: Visualization of the correlation attention matrix on COVID-19. Left: CasMLN. Right: our method.

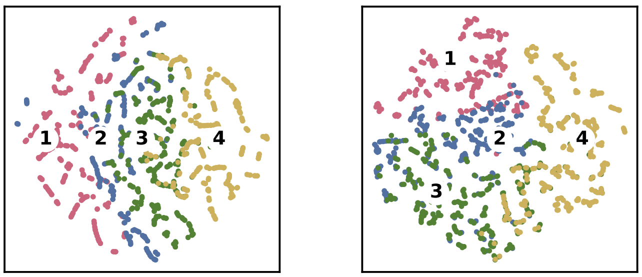

Figure 10: t-SNE visualization of embeddings on COVID-19. Left: CasMLN. Right: our method.

and mean of case counts. Regions with higher case counts generally correspond to denser populations and stronger human mobility. As shown in Figure 8, our method consistently achieves lower prediction errors than the strongest baselines, with the largest gains appearing in high-mobility regions and severe-outbreak periods.

This robustness is further supported by the hidden-layer correlation attention patterns in Figure 9. Compared with CasMLN, our method yields a clearer and more structured correlation matrix, with stronger associations among epidemiologically related states and less diffuse noise among irrelevant nodes. This suggests that unified latent relation modeling enhances effective inter-node dependencies rather than uniformly increasing all pairwise correlations, enabling better modeling of cross-state transmission, temporal lag effects, and long-range spatial-temporal dependencies under heterogeneous outbreak intensities.

This finding is consistent with Figure 10, where our method produces more compact and discriminative node representations than existing SOTA methods. Together, these results indicate that the superior robustness of our method stems from learning structured and semantically meaningful node interactions in the hidden attention space, which ultimately improves prediction accuracy on complex epidemic dynamics.

5.8 Few-Shot and Zero-Shot Prediction

Tables 4, 5, 6, and 7 report the few-shot and zero-shot prediction results on Mine Pressure and COVID-19 with different LLM backbones. Overall, our method consistently outperforms all baselines under both few-shot and zero-shot settings across different datasets and backbones, demonstrating strong generalization ability and data efficiency.

Table 4: Few-shot prediction results on Mine Pressure with different LLM backbones.

Method	GPT-2		LLaMA-3.2		LLaMA-2		LLaMA-3.1	
	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE
OFA	3.78	6.42	3.65	6.18	3.52	6.05	3.60	6.12
GATGPT	3.22	5.60	3.16	5.52	3.10	5.45	3.08	5.32
GCNGPT	3.19	5.58	3.14	5.50	3.08	5.12	3.05	5.40
ST-LLM	3.12	5.48	3.08	5.40	3.03	5.35	3.30	4.60
ST-LLM+	3.05	5.40	3.04	5.30	2.95	5.25	2.92	5.20
Our	**1.90**	**3.30**	**1.93**	**3.35**	**1.85**	**3.25**	**1.82**	**3.22**
Imp. (%)	37.70	38.89	36.51	36.79	37.28	36.52	37.67	35.96

Table 5: Few-shot prediction results on COVID-19 with different LLM backbones.

Method	GPT-2		LLaMA-3.2		LLaMA-2		LLaMA-3.1	
	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE
HTG-LLM	690	1520	685	1508	675	1485	668	1470
RGAT-LLM	620	1350	615	1338	605	1315	598	1302
RGCN-LLM	645	1400	640	1388	630	1365	622	1350
Our	**530**	**1120**	**525**	**1110**	**515**	**1090**	**508**	**1080**
Imp. (%)	14.52	17.03	14.63	17.04	14.87	17.11	15.05	17.05

Table 6: Zero-shot prediction results on Mine Pressure with different LLM backbones.

Method	GPT-2		LLaMA-3.2		LLaMA-2		LLaMA-3.1	
	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE
OFA	3.95	6.75	3.88	6.40	3.77	6.35	3.83	6.40
GATGPT	3.43	6.03	3.46	6.08	3.35	5.93	3.31	5.88
GCNGPT	3.40	6.00	3.43	6.05	3.33	5.90	3.28	5.85
ST-LLM	3.35	5.95	3.38	6.00	3.28	5.85	3.23	5.80
ST-LLM+	3.33	5.93	3.36	5.98	3.25	5.83	3.20	5.78
Our	**2.05**	**3.55**	**2.08**	**3.60**	**2.00**	**3.50**	**1.97**	**3.47**
Imp. (%)	38.44	40.13	38.10	39.80	38.46	39.97	38.44	39.96

Table 7: Zero-shot prediction results on COVID-19 with different LLM backbones.

Method	GPT-2		LLaMA-3.2		LLaMA-2		LLaMA-3.1	
	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE
HTG-LLM	750	1650	745	1638	735	1615	728	1600
RGAT-LLM	680	1507	675	1488	665	1465	658	1450
RGCN-LLM	705	1560	700	1548	690	1525	682	1510
Our	**575**	**1215**	**570**	**1205**	**560**	**1185**	**552**	**1175**
Imp. (%)	15.44	19.38	15.56	19.02	15.79	19.11	15.08	18.97

6 Conclusion

In this work, we propose a unified framework for spatiotemporal graph representation learning with large language models. Its core Latent Relation Adapter, composed of an LR-Unifier and an LR^2-Former, unifies cross-time spatial relations and performs structured local-to-global relational aggregation. Experiments on six real-world datasets demonstrate that our method is effective and robust for node regression and link prediction.

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